

Carolina Del Pilar Díaz

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Summary

AI/ML Engineer, Senior Data Scientist, and Applied Statistician (Economist) with 20+ years of experience operating as a Strategic AI/ML Consultant & Forward Deployed Engineer (FDE) for global enterprises. I specialize in architecting end-to-end AI products and solutions using Python, Machine Learning, and Deep Learning. I build mathematically robust Multi-Agent Systems (GenAI, RAG, MCP) that prevent model hallucinations in regulated sectors. I have architected and shipped full-stack AI applications, combining the power of Python with the rigor of Advanced Statistics, Causal Inference, and MLOps on on-premises, hybrid, and cloud infrastructures across Fintech, AgTech, and Healthcare.

Experience

IT Patagonia (Consultant FDE), Senior AI Engineer

Buenos Aires, Argentina

June 2024 – present

- Forward Deployed Engineer (FDE) & Technical Lead: Directed the architecture and deployment of Generative AI, RAG, and Multi-Agent solutions for enterprise clients (Banco Galicia, ICBC), defining technical roadmaps and optimizing cloud infrastructure.
- Agentic AI & Complex Workflows: Architected a 33-node LangGraph state-machine for automated legacy codebase documentation (Patagon.IA) and engineered a 100% asynchronous LlamaIndex RAG pipeline with parallel retrieval for a real-time sales assistant (Facturante).
- LLM Cost Optimization & Resiliency: Implemented state-checkpointing to avoid API reprocessing in long-running LLM pipelines, enforced strict token budgets (ThinkingConfig), and designed dynamic routing between cloud (AWS Bedrock) and local (Ollama) foundational models.
- Quality Assurance & Serverless Deployments: Built an advanced 'LLM-as-a-judge' multi-evaluator system using majority voting to eliminate hallucinations. Shipped containerized (Docker) hybrid deployments and serverless MLflow PyFunc inference models on Databricks.

ITP Patagonia - ICBC, Senior Data Scientist

Buenos Aires, Argentina

June 2024 – July 2025

- Developed predictive models for payroll clients and mutual fund investment using AWS SageMaker, AutoML, Canvas, and Neural Network and XGBoost.
- Built an anomaly detection model using Neural Networks (AENN), Isolation Forest, and SHAP for interpretability using AZURE framework.
- Currently developing a fraud detection model with AWS Fraud Detector, leveraging Decision Trees, Neural Networks, Time Series Analysis, and Custom Fraud Prediction using AutoML.

Nubiral S.A., Senior Data Scientist. COE, Commission of Expert. Generative Models.

Buenos Aires, Argentina

Oct 2023 – May 2024

- Airline solution: Integrated AWS Rekognition, Lambda, S3, LangChain, and OpenAI GPT for Computer Vision and Generative AI, improving object detection accuracy at Aerolineas Argentina.
- Developed a Post-Call Analytics solution using AWS services (Transcribe, Comprehend, Lambda, QuickSight) to analyze customer interactions.
- Implemented Azure Cognitive Search and Form Recognizer for efficient PDF document management at Banco Macro.

Pi Data Strategy & Consulting , Semi-Senior Data Scientist, Statistician, Economist		Córdoba, Argentina
• Creating data regression models, predictive data modelling, and analyzing data mining algorithms. Modelling along creating end to end pipeline to deploy. Azure Technology. • Achieve EDA, Statistical analysis, regression models, classification models, neuronal network. • Sales Forecast for georeferenced Dataset. Multinomial predictive models applied to the level of customer satisfaction.		Mar 2021 – Oct 2023
National Institute of Agriculture Technology (INTA) , Lead Statistician / Head of Statistical Research		Manfredi, Córdoba, Argentina
• Designed and implemented experimental designs for all research projects, ensuring robust statistical methodology across multiple areas. • Applied neural networks, deep learning, and computer vision for UAV and agricultural machinery, enhancing crop health monitoring.		Jan 2012 – Dec 2021

Education

Postgraduate Master's Degree	National University of Córdoba , Applied Statistics	Córdoba, Argentina Jan 2006 – Dec 2008
Diploma	National University of Córdoba , Data Science Machine Learning and applications	Córdoba, Argentina Jan 2019 – Dec 2020
Bachelor's degree	National University of Córdoba , Economics	Córdoba, Argentina Jan 1993 – Dec 1998

Skills

Agentic & Generative AI: Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, AutoGen, Semantic Kernel, RAG, MCP, Prompt Engineering, LLMs (Gemini, GPT, Claude, Ollama, Hugging Face)

Machine Learning & Stats: Deep Learning, Computer Vision, Causal Inference, Bayesian Statistics, Econometrics, Anomaly Detection, Time Series, MLOps

Cloud & Big Data: AWS (SageMaker, Lambda, Rekognition, Fraud Detector), Azure AI Services, Snowflake, Databricks, Apache Spark, Python, R, SAS

Languages: Spanish (Native), English (Intermediate), Portuguese (Basic)

Projects

PATAGONIA — Automated COBOL Legacy Documentation	Jan 2026
33-node LangGraph pipeline on AWS Bedrock generating BRD/TRD docs from COBOL source. Led prompt engineering (~23 sections), a 3-judge LLM verification panel (majority voting) and anti-hallucination filters vs. a deterministic golden-truth. ITPCS — 20/20 sections verified, 0 pipeline errors. IT Patagonia.	
Banco Galicia — AI Architecture & Observability Assessment	Jan 2026
Code-based audit of two production AI systems (Java/Kafka/OpenShift multi-agent orchestrator + Azure OpenAI RAG). Designed the observability strategy (Prometheus, Grafana, OpenTelemetry/Jaeger, RAGAS) and a target cloud-native architecture aligned to BCRA regulation. IT Patagonia.	
Banco Macro — Nightly Batch Completion Predictor (OPTI+)	2025-2026
Statistical predictor of batch end-time as an hour window. Compared Linear Regression, Random Forest, HistGradientBoosting and Prophet (MAE/R ²) plus an instance-based k-NN estimator with robust median and asymmetric ranges. Leave-one-out over 105 days — 99% within SLA. FastAPI + React + Docker, live OPTI API. IT Patagonia.	

ICBC — Banking Customer Segmentation (RFM) RFM segmentation on AWS SageMaker Canvas/Data Wrangler over Amazon Athena (customer-360 join, ~250 features). Feature engineering with robust-σ outlier handling, imputation, ordinal/one-hot/similarity (MinHash) encoding and STL/LOESS decomposition over ~1M monthly records. IT Patagonia.	Jan 2025
Facturante — WhatsApp Sales AI Agent Conversational sales agent deployed as an MLflow PyFunc model on Databricks Model Serving with RAG (Databricks Vector Search) and Gemini. Automated LLM-as-judge evaluation over a golden dataset, consistency scoring and Lakeview conversation auditing. IT Patagonia.	Jan 2026
ITP Flokzu — Agentic Document Platform Two-agent platform on AWS Lambda + Bedrock (Claude Sonnet/Haiku, Strands framework) for invoice field extraction and Knowledge-Economy eligibility decisions, integrated with Flokzu BPM; CloudWatch monitoring. IT Patagonia.	Jan 2026
ICBC — Legacy Modernization VB6 → C#/.NET 8 Migration of the judicial-orders banking system (AFIP/BCRA integration) from VB6 to a layered .NET 8 architecture with EF Core, Dependency Injection, async/await, Serilog and DB2. IT Patagonia.	Jan 2026
Interbanking — Conversational Operations Agent (POC) Technical proposal for a governed SQL + RAG agent on Databricks + S3 (Lakehouse Medalion, Unity Catalog RLS/CLS, Model Serving, Vector Search, grounded RAG). IT Patagonia.	Jan 2026
Universal Assistance — QA Test Automation Web app generating Playwright (TypeScript) test scripts from natural-language requirements using Gemini. React 19 + Vite. IT Patagonia.	Jan 2026
Banco Macro — Intelligent Assets Platform Web app for asset-data analysis with conversational AI (Gemini), automatic column mapping, Recharts dashboards and a 3-month cost forecast. IT Patagonia.	Jan 2026
Country Expansion Program Demographic segmentation. Forecast the Future Demand for actual cities. Anheuser-Busch InBev. Pi consulting & Strategies.	Jan 2022
Logistics prediction of the level of customer satisfaction Health Score for NPS. Anheuser-Busch InBev. Pi consulting & Strategies.	Jan 2022
Genotype by Environmental Predictions GDM. Don Mario. Data Science. Pi consulting & Strategies.	Jan 2021
Digital Government Project of the Ministry of Economy and Energy Data Science, Natural Language Process. Pi consulting & Strategies.	Jan 2021
Text Classifications to improve the management system Pernod Ricard. Data Science. Pi consulting & Strategies.	Jan 2021
PNCYO 1127046 Evaluations networks unpublished of germplasm INTA. Statistical Analysis of Genotypes regarding environments. Publications of the results. INTA.	2013-2019

PNBIO 1131042 Applied Genomics to molecular optimization parametrical and non-parametrical statistical models. INTA.	2013-2019
PNBIO 1131043 Bioinformatics and genomics. Assistance and new models in the evaluation of big base data. INTA.	2013-2019
CORDO 1262206 Transfer and training in the application in new experimental designs. INTA.	2013-2019
CORDO 1262205 Technology in both extensive and intensive crops under irrigation and rainfed. INTA.	2013-2019

Publications

The bright side of alfalfa viral diseases: The response of coumestrol phytoestrogen content in alfalfa genotypes Pilar Silva, M., Basigalup, D. H., <i>Carolina Del Pilar Díaz</i> (Plant Pathology)	Jan 2020
Segregations Analysis of Polymorphic SSR in three populations of F2 of peanut for charcoal tolerance Ritta L., Moreno M.V., Mamaní E., <i>Díaz C.</i> , Etchart V. (Peanut Workday)	Jan 2020
Advanced Mixed Generalized linear models, for effects of two levels of residues of Suico extracts on ruminal fermentation in vitro <i>Díaz, C. P.</i> , Scorcione Turcato, M.C, Ortiz Chura A. (XXIII Reunión Científica del GAB)	Jan 2019
The influence of insecticide applications on stored soybeans on the level of residues in the crude oil Aguilar Roxana, Nassetta Mirtha, <i>Díaz Carolina</i> , Rojas Dante (7th Congress International Food Science and Technology)	Jan 2019
Nutritional chemical composition of fruits of different carob species from northwestern Córdoba Martínez MJ, Ledesma, M, Aguilar R, <i>Díaz C.</i> , Nassetta M (VII International Congress of Food Sciences and Technology)	Jan 2019
Assessing INTA's Arachis hypogaea Core Collection for reaction to peanut smut Mamaní EMC, Rodríguez AV, Cordes G, <i>Díaz CDP</i> , Moreno MV (Conferencia del Genoma del maní)	Jan 2018
Corn Networks: Analysis of genotype-environment. Application of GGE ShuklaModels <i>Díaz Carolina Del Pilar</i> , Ferreyra M.Laura (Horizonte magazine)	Jan 2018
Alfalfa (Medicago sativa L.) coumestrol phytoestrogen content in response to genotypes, cultivar and viral disease Silva, M., Chiacchiera, S., Mamaní, E., <i>Díaz, C.</i> (Second World Alfalfa Congress)	Jan 2018

Regional Evaluation of INTA Wheat Varieties Cordes D., Candela R., <i>Díaz C.</i> (Rural exhibition of Jesús María)	Jan 2018
Cluster and Yield Evaluation for Corn Varieties Chincuini D., Druetta A., <i>Díaz C.</i> , Vega C. (Rural exhibition of Jesús María)	Jan 2018
Phenotyping for Drought Tolerance in Sunflower inbred lines Grandon N., Martin E., Cicconi E., <i>Díaz C. P.</i> (32nd Meeting of Plant Physiology Congress)	Jan 2018
Effect of Chemical and Biological control of acid vine rot Cordes, D.D., Grion, A., <i>Díaz C.</i> (40th Argentine Congress of Horticulture)	Jan 2018
Effect of Lippia turbinata G extracts, at two levels of inclusion, on in vitro ruminal fermentation Scorcione Turcato, M.C., Ortiz Chura A., <i>Díaz, C. D. P.</i> (41° Congreso de la Asoc. Argentina de Producción Animal)	Jan 2018
Deltamethrin residues in crude soybean oil Aguilar, R., <i>Díaz, C.</i> , Strada, J. (Periurban Conferences)	Jan 2017
Analysis of Chlorpyrifos methyl residues in crude soybean oil from treated grains in storage Aguilar R., Nassetta M., <i>Díaz C.</i> (IV Ibero-American Congress of Environmental Health)	Jan 2017
Dissipation dynamics of organophosphate insecticides in crude oil of soybean Aguilar R., <i>Díaz C.</i> , Strada J., Nassetta M. (V International Congress of Food Science and Technology)	Jan 2016
Effects of the alfalfa stunt viral complex on different genotypes estimated with Mixed Linear Models <i>Díaz, C.</i> , Silva, M., Mamaní, E., Basigalup, D. (XXI GAB Scientific Meeting)	Jan 2016
Genetic characterization of cultivated peanut genetic resources using microsatellite markers Pozzi, F, Etchart V, <i>Díaz C.</i> , Moreno, M. V (Rev. FCA UNCUYO)	Jan 2014

Certificates and Courses

- Agentic AI on AWS. BeSA- Become a Solution Architect. Candidate. (2026)
- Databricks Certified Generative AI Engineer Associate (2026)
- Microsoft Certified: Azure AI Fundamentals. Candidate. (2026)
- AWS Academy Cloud Architecting – Candidate, WIC Training Program (2025)
- AWS Academy Graduate - Cloud Foundations (2024)
- Generative AI Large Language Models & ChatGPT Prompt Engineering for Developers. Deep Learning AI. (2023)
- MLOps Fundamentals CI/CD/CT pipelines with Azure (2022)
- Databricks and Apache Spark for Big Data (2022)
- Machine Learning Pipelines with Azure ML Studio & Azure Synapse SQL Pool (2021)
- Bayesian Statistics: Techniques and Models. University of California (2021)
- AWS Academy Graduate - Machine Learning (2020)
- Diploma in Data Science Machine Learning and applications, National University of Córdoba (2020)
- Machine Learning by Professor Andrew Ng, Stanford University / Coursera (2018)
- Brief Course on Machine Learning & Brief Course on Mixed Models (2016)
- Post-Graduate Course of Bayesian Statistics, National University of Córdoba (2016)